Smart Wallet

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Abstract :

As smart features are added to commonplace items, users' convenience and security are improved. This is a trend in the field of contemporary technology. A unique idea is presented in this abstract: a smart wallet with sophisticated sensors that can determine whether it is open or closed and whether cards are inside. By strategically embedding Piezoelectric sensor within its structure, the suggested smart wallet detects whether sensor is conducting or not. The user can receive immediate feedback through smartphone applications thanks to these sensors, which allow for real-time monitoring of the wallet's physical condition.

Index Terms: Piezoelectric Sensor,Conductors,Esp32-wifi module.

1 Introduction

In the era of rapid technological advancement, the convergence of traditional accessories with smart capabilities has revolutionized the way we interact with everyday objects. One such innovation is the Smart Wallet, a modernized version of the conventional wallet equipped with sophisticated sensors and conductive elements.Traditionally, wallets have served as simple repositories for cash, cards, and identification, offering limited functionality beyond basic storage. However, the integration of piezoelectric sensors and conductors introduces a new dimension of intelligence to this essential accessory. Piezoelectric sensors, known for their ability to generate electrical charge in response to mechanical stress, are strategically positioned within the wallet's framework.

2 Background

Smart wallets available today typically integrate various technological features to enhance convenience, security, and functionality for users. Here's a brief overview of some common features found in our smart wallets:

- RFID Blocking: Many smart wallets incorporate RFID-blocking technology to protect users' contactless cards from unauthorized scanning or skimming.

- Bluetooth Connectivity: Some smart wallets are equipped with Bluetooth connectivity, allowing them to pair with smartphones or other devices. This enables features such as GPS tracking to locate a misplaced wallet, as well as notifications to alert users if they leave their wallet behind.

- Integrated Trackers: Some smart wallets include built-in trackers powered by GPS or Bluetooth technology, enabling users to locate their wallet using a smartphone app in case it's lost or stolen.

- Charging Capability: Certain smart wallets come with built-in power banks or wireless charging capabilities to recharge smartphones or other devices on the go.

- Material Durability: Smart wallets often utilize durable materials such as carbon fiber, aluminum, or leather to ensure longevity and protection against wear and tear.

3 Competitor

Let's delve into brief descriptions of some key competitors in the smart wallet market:

- Ekster

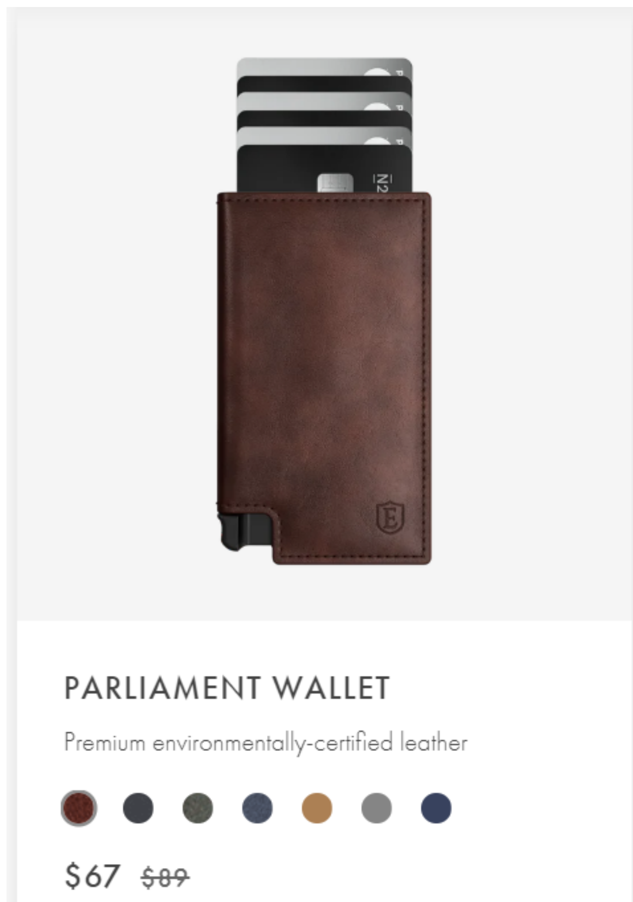

Figure 1. Ekster smart wallet

Ekster is renowned for its sleek and minimalist smart wallets. They often feature RFID-blocking technology to prevent unauthorized scanning of contactless cards. Ekster wallets also integrate a card ejection mechanism, allowing users to access their cards with a push of a button.

- Secrid:

Figure 2. Secrid smart wallet

Secrid is known for its innovative Cardprotector wallets, which offer RFID protection in a slim and durable aluminum casing. These wallets prioritize functionality and security, allowing users to carry their essential cards while protecting them from electronic theft.

- Arista Vault:

Arista Vault is an Indian brand which was featured in the recent Shark tank India for their product funding.These wallet provide RF-ID card detection and track the wallet.

The smart wallets currently available in the market tend to be priced quite high, making them inaccessible to many people. Our aim is to offer a smart wallet at a much more affordable price point, ensuring that everyone can afford to purchase one and enjoy the convenience it provides.

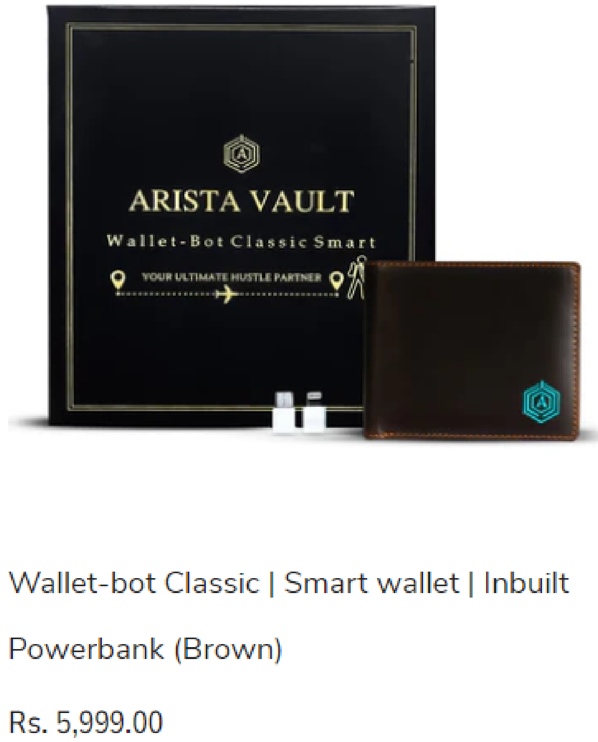

Figure 3. Arista Vault smart wallet

4 Working

- **Point 1: Description of the piezoelectric sensor setup**
  For our product, we've implemented a unique approach using piezoelectric sensors as conductors. Two piezoelectric sensors are employed: one receives a 5V analog input, while the other serves to capture the output signal. When both conductors are in contact, the output registers as 5V; when they're not in contact, the output drops to 0V. This setup allows us to establish a threshold: if the output reads 5V, indicating no card presence, and conversely, if it's 0V, suggesting a card is present in the section.
- **Point 2: Dual functionality of the piezoelectric sensors**
  Moreover, this configuration isn't limited to card detection; it also serves to determine whether the wallet is open or closed. A 5V output signifies that the wallet is closed, while a 0V output indicates that it's open. By leveraging this dual functionality of the piezoelectric sensors, we provide users with a comprehensive solution that not only detects card presence but also monitors the status of the wallet, ensuring enhanced security and convenience.
- **Point 3: Transmission of data to ESP32 WiFi module**
  By incorporating a greater number of slots, our system can detect the presence of multiple cards simultaneously, thereby enhancing its capacity for card detection. The collected data is then transmitted to an ESP32 WiFi module. This module serves as the intermediary, relaying the information to either a dedicated application or a designated website, providing real-time updates on the status of the wallet and the number of cards detected.
- **Point 4: Scalability and customization**

Expanding on this capability, the system offers scalability, allowing users to customize the configuration based on their specific needs. Whether it's a few essential cards or a multitude of items, our solution accommodates varying requirements with ease.

- **Point 5: Connectivity and remote monitoring**
  Furthermore, the integration of the ESP32 WiFi module enables seamless connectivity, facilitating instant notifications and remote monitoring. Users can stay informed about the status of their wallet and its contents, regardless of their location, fostering a sense of security and peace of mind.
- **Point 6: Summary statement**
  In essence, our system combines versatility, connectivity, and convenience, empowering users with a comprehensive solution for managing their wallet and card inventory efficiently.

5 Results

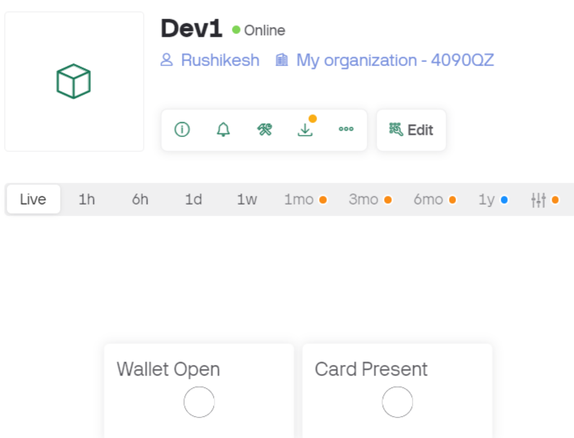

Figure 4. Wallet is closed and card is not present

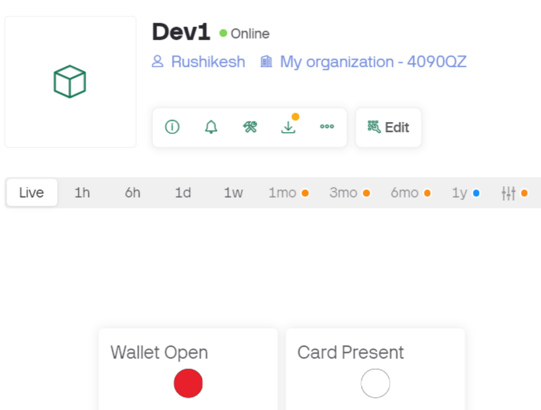

Figure 5. Wallet is open and card not present

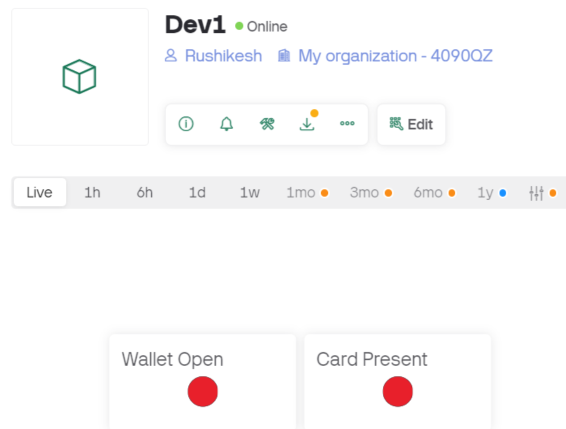

Figure 6. Wallet is open and card is also present

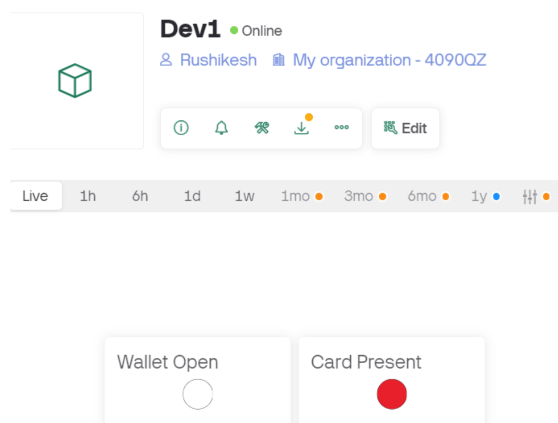

Figure 7. Wallet is closed and card is present

6 Conclusion

In summary, the Smart Wallet presented herein represents a paradigm shift in the evolution of traditional accessories, blending advanced sensor technology with everyday functionality. By harnessing the power of piezoelectric sensors and conductors, this intelligent wallet offers a seamless and intuitive user experience while prioritizing security and peace of mind.

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